Rundong Luo

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Education

Cornell University — Ithaca, NY
Ph.D. in Computer Science — Aug. 2024 – Present
Advisors: Prof. Wei-Chiu Ma and Prof. Noah Snavely

Peking University — Beijing, China
Bachelor of Science in Computer Science and Technology (summa cum laude) — Aug. 2020 – Jun. 2024
Advisors: Prof. Yisen Wang and Prof. Jiaying Liu

Publications

* denotes equal contribution.

- **Rundong Luo**, Yunong Liu*, Di Cao*, Matthew Tancik, Shyamal Buch, Colton Stearns, and Wenqi Xian. SceneRig: An Agentic System for Simulation-Ready 3D Scene Reconstruction from Single Images. **[arXiv 2026]** [arXiv] [project]
- Guangzhao He*, **Rundong Luo***, Wei-Chiu Ma, and Hadar Averbuch-Elor. Thinking in Blender: Staged Executable Inverse Graphics with Vision-Language Models. **[arXiv 2026]** [arXiv]
- Haozheng Yu, Xinyu Yang, **Rundong Luo**, Jennifer J. Sun, and Bharath Hariharan. SMG: Semantic Motion Graph for Monocular Dynamic Gaussian Splatting. **[ECCV 2026]** [arXiv] [project]
- Yen-Siang Wu, **Rundong Luo**, Jingsen Zhu, Tao Tu, Ali Farhadi, Matthew Wallingford, Yu-Chiang Frank Wang, Steve Marschner, and Wei-Chiu Ma. Seeing Fast and Slow: Learning the Flow of Time in Videos. **[ECCV 2026]** [arXiv] [project]
- **Rundong Luo**, Noah Snavely, and Wei-Chiu Ma. ShadowDraw: From Any Object to Shadow–Drawing Compositional Art. **[CVPR 2026]** [arXiv] [project]
- **Rundong Luo**, Matthew Wallingford, Ali Farhadi, Noah Snavely, and Wei-Chiu Ma. Beyond the Frame: Generating 360° Panoramic Videos from Perspective Videos. **[ICCV 2025]** [arXiv] [project]
- **Rundong Luo***, Haoran Geng*, Congyue Deng, Puhao Li, Zan Wang, Baoxiong Jia, Leonidas Guibas, and Siyuan Huang. PhysPart: Physically Plausible Part Completion for Interactable Objects. **[ICRA 2025]** [arXiv] [project]
- **Rundong Luo***, Hong-Xing Yu*, and Jiajun Wu. Unsupervised Discovery of Object-Centric Neural Fields. **[TMLR 2025]** [arXiv] [project]
- Wenjing Wang*, **Rundong Luo***, Wenhan Yang, and Jiaying Liu. Unsupervised Illumination Adaptation for Low-Light Vision. **[TPAMI 2024]** [project]
- **Rundong Luo**, Wenjing Wang, Wenhan Yang, and Jiaying Liu. Similarity Min-Max: Zero-shot Day-Night Domain Adaptation. **[ICCV 2023 (Oral)]** [arXiv] [project]
- **Rundong Luo***, Yifei Wang*, and Yisen Wang. Rethinking the Effect of Data Augmentation in Adversarial Contrastive Learning. **[ICLR 2023]** [arXiv]

Work Experience

Luma AI — May 2026 – Present
Research Scientist Intern | Mentors: Dr. Wenqi Xian and Dr. Colton Stearns — Redwood City, CA
- Developed SceneRig, an agentic system for simulation-ready 3D scene reconstruction from single images, combining object-centric reconstruction with geometric and physical feedback. Validated reconstructed scenes through real-to-sim policy evaluation and robot trajectory replay.

Research Experience

Cornell Vision and Graphics Lab — Aug. 2024 – Present
Advisors: Prof. Wei-Chiu Ma and Prof. Noah Snavely — Cornell University
- Conduct research in 3D computer vision, inverse graphics, and video generation.
- Developed methods for generating 360° panoramas from perspective videos (ICCV 2025) and learning temporal transformations in video (ECCV 2026).
- Created a computational system that turns arbitrary objects into compositional shadow art (CVPR 2026).

Stanford Vision and Learning Lab — Jan. 2023 – Oct. 2023
Advisor: Prof. Jiajun Wu — Stanford University
- Developed an unsupervised method for discovering object-centric neural fields in 3D scenes (TMLR 2025).

STRUCT Lab — Apr. 2022 – Jun. 2024
Advisor: Prof. Jiaying Liu — Peking University
- Studied improving computer vision systems in low-light environments. Published first-author papers in ICCV 2023 (oral) and TPAMI.

ZERO Lab — Jul. 2021 – Sept. 2022
Advisor: Prof. Yisen Wang — Peking University
- Studied data augmentation for adversarial contrastive learning and published a co-first-author paper (ICLR 2023).

Academic Service

- **Workshop Organizer:** Synthetic Data in Computer Vision, CVPR (2025–2026).
- **Conference Reviewer:** CVPR and NeurIPS (2024–2026); ICLR (2025–2026); WACV, ICCV, and ICML (2025); ECCV, 3DV, SIGGRAPH, and SIGGRAPH Asia (2026).
- **Teaching Assistant, Cornell:** Foundations of Robotics (Fall 2026); Introduction to Computer Vision (Spring 2026); 3D Vision (Fall 2024).
- **Teaching Assistant, Peking University:** Practice of Programming in C&C++ (Spring 2023).

Selected Honors and Awards

- SenseTime Scholarship (awarded to 30 undergraduates in AI-related major per year nationwide), 2024.
- Outstanding Undergraduate Thesis of Beijing (190 recipients among 4000+ at Peking University), 2024.
- Outstanding Graduate of Beijing (34 recipients among 4000+ at Peking University), 2024.
- Chinese National Scholarship (awarded to 0.2% of undergraduates in China), 2023.